

Data Driven Approaches in (Multi-objective) Bayesian Optimisation - An Introductory Course

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In this course

You will learn

- Why multi-objective optimisation
- What are existing multi-objective optimisation methods and their shortcomings
- Introduction to Bayesian machine learning - Gaussian Processes
- Why Bayesian (multi-objective) optimisation
- How to utilise Bayesian inference to alleviate computation time
- Examples of real-world optimisation problems

Course Structure

Day 1

- Introduction to multi-objective optimisation
 - Definition of multi-objective optimisation problems (MOPs)
 - Pareto dominance and Pareto optimality

Course Structure

Day 2

- Introduction to Gaussian Processes as Machine learning models
 - Training Gaussian processes
 - Prediction with Gaussian processes

Course Structure

Day 3

- Bayesian Optimisation
 - Exploitation vs Exploration
 - Acquisition functions

Course Structure

Day 4

- Scalarising functions
 - Role of Decision-Maker
 - A priori, a posteriori and interactive methods

Course Structure

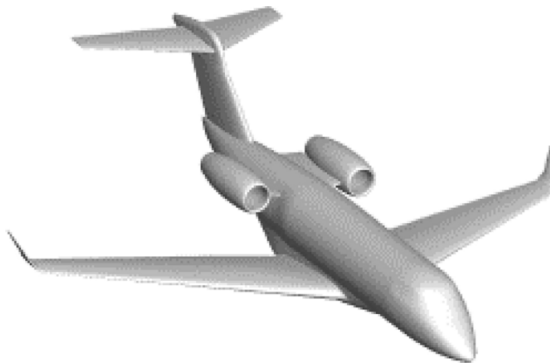
Day 5

- Multi-objective Bayesian Optimisation
 - Mono-surrogate approaches
 - Multi-surrogate approaches
 - Summary

Any questions on the module structure?

Examples of Optimisation

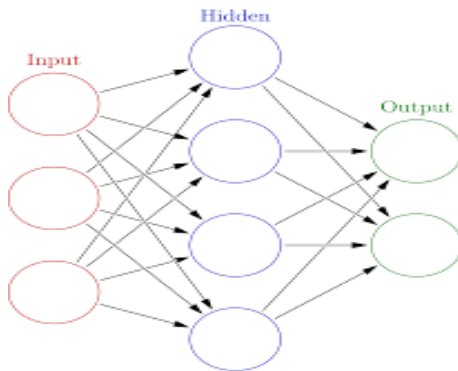
Engineering Design¹



¹Forrester, A. I. J., Sobester, A., & Keane, A. J. (2008). Engineering Design via Surrogate Modelling: A Practical Guide. Wiley

Examples of Optimisation

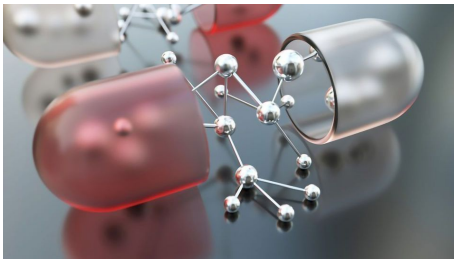
Hyperparameter tuning in Neural Networks²



²Wu et al. Hyperparameter Optimization for Machine Learning Models Based on Bayesian Optimization, Journal of Electronic Science and Technology, 17, 2019, 26-40.

Examples of Optimisation

Drug Discovery³



³Colliandre L, Muller C. Bayesian Optimization in Drug Discovery. Methods Mol Biol. 2024;2716:101-136.

Examples of Optimisation

- Material Science⁴
- Financial Modelling and Portfolio Optimisation⁵
- Reinforcement learning⁶

⁴Xue, D., Balachandran, P., Hogden, J. et al. Accelerated search for materials with targeted properties by adaptive design. Nat Commun 7, 11241 (2016).

⁵Gupta, A. A Bayesian Approach to Financial Model Calibration, Uncertainty Measures and Optimal Hedging. Oxford University, UK, 2010.

⁶Paul et al. , JMLR, Robust Reinforcement Learning with Bayesian Optimisation and Quadrature, JMLR, 21, 1-31, 2020.

Terminology

Decision vector

$\mathbf{x}$ is a decision vector (a design), $\mathbf{x} = (x_1, x_2, \dots, x_D)^\top$

$\mathcal{X}$ is the decision space, where decision vectors satisfying all problem constraints reside

Here, without loss of generality (wlog), we will be illustrating on continuous spaces, $\mathcal{X} \subset \mathbb{R}^D$

Constraints

In many situations, $\mathcal{X}$ is defined by **box constraints** on the continuous space

- E.g., x_1 is the diameter of a pipe in an engine, and cannot be lower than 0.1cm or larger than 3cm

Taken together, all the upper and lower bounds ‘cut out’ a hyperrectangle from the continuous space, every point in the rectangle is legal (in $\mathcal{X}$), everything outside is illegal, and not a valid decision vector **[example on board]**

Constraints

When the constraints are box constraints it is relatively easy to draw a member of $\mathcal{X}$, and check legality of a putative decision vector

Other more complicated constraint forms can exist, typically of the form $g_i(\mathbf{x}) \leq$ and $h_i(\mathbf{x}) = 0$ **[examples on board]**

For some optimisation problems, there high number and complexity of constraints means the the task is simply to find **any** member of $\mathcal{X}$ (e.g. timetabling/scheduling problems)

Objective function

Alongside the legal domain, is the objective function f . When the value returned is to be minimised, this is often also referred to as a **cost function**

Using the objective function we access the **quality** of a decision vector, $y = f(\mathbf{x})$

Note it is possible for the function to require other arguments/parameters $y = f(\mathbf{x}, \theta)$, where we do not have control (in the optimisation) over θ

The single objective optimisation task

Focusing, wlog, on minimisation, we seek:

$$\mathbf{x}^* = \operatorname{argmin}_{\mathbf{x} \in \mathcal{X}} f(\mathbf{x})$$

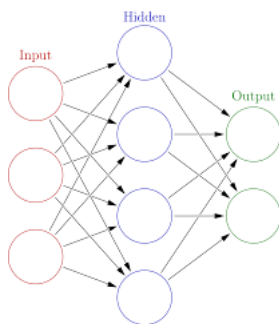
Solving Single Objective Optimisation Task

We can put an **order** members of $\mathcal{X}$:

- if $f(\mathbf{x}) < f(\mathbf{x}')$, then $\mathbf{x}$ is better than $\mathbf{x}'$
- if $f(\mathbf{x}) > f(\mathbf{x}')$, then $\mathbf{x}$ is worse than $\mathbf{x}'$
- if $f(\mathbf{x}) = f(\mathbf{x}')$, then $\mathbf{x}$ is equivalent to $\mathbf{x}'$

These observations seem trivial — however, as we shall see, the situation is not as clean cut in the multi-objective setting

Examples Single Objective Optimisation Task



Training a neural network requires optimising a loss function:

$$L(y, \hat{y}) = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2$$

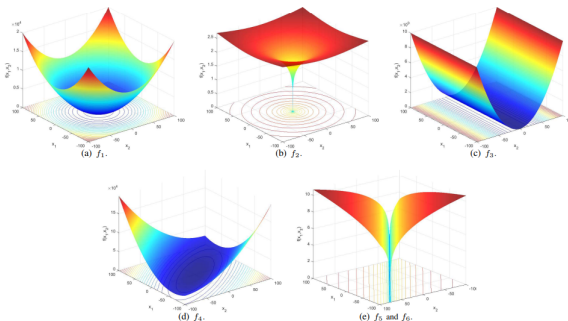
Examples Single Objective Optimisation Task



Travelling Salesman Problem: Given a list of cities and the distances between each pair of cities, what is the shortest possible route that visits each city exactly once and returns to the origin city?

Examples Single Objective Optimisation Task

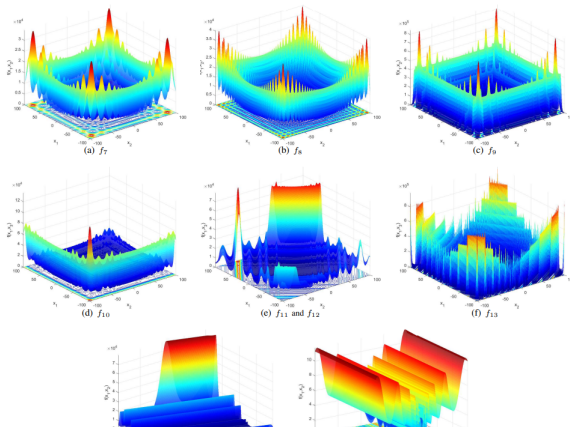
Benchmark problems - test of algorithms



Source; GNBG-Generated Test Suite for Box-Constrained Numerical Global Optimization <https://arxiv.org/pdf/2312.07034>

Examples Single Objective Optimisation Task

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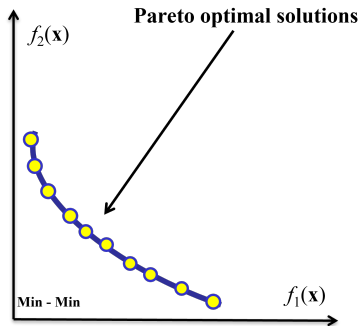
Multi-objective Optimisation

Multiple objective functions:

$$\text{minimise } f(\mathbf{x}) = (f_1(\mathbf{x}), \dots, f_M(\mathbf{x})) \quad \text{subject to } \mathbf{x} \in \mathcal{X}$$

Example

- Minimising manufacturing cost and maximising the quality of product
- Maximising the fuel efficiency and minimising the soot emission and minimising NOx



Multi-objective Optimisation

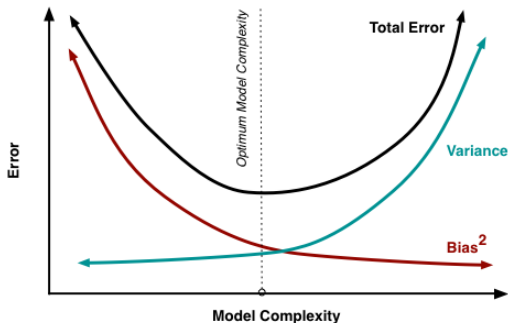
- In multi-objective optimisation, we have $M > 1$ objectives
- The **objective vector** y is returned as $y = f(x)$
- A putative solution is the tuple of a valid design and its evaluation:
 $s = (y, x)$

The MOO Task

$$x^* = \operatorname{argmin}_{x \in \mathcal{X}} f(x)$$

Examples Multi-objective Optimisation

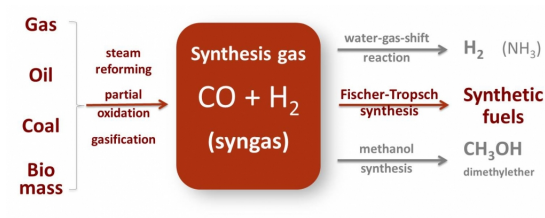
Bias and Variance Trade off



$$\text{Bias: } E(\hat{y}) - y$$
$$\text{Variance: } E[(\hat{y} - E[\hat{y}])^2]$$

Examples Multi-objective Optimisation

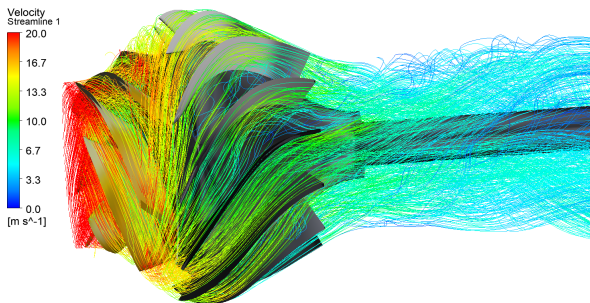
Chemical Engineering: Synthesis gas production



Objectives: 1. Maximise CH_4 rate 2. Maximise CO rate, and 3. Minimise H to CO ratio

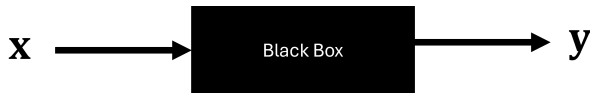
Examples Multi-objective Optimisation

Pump shape optimisation



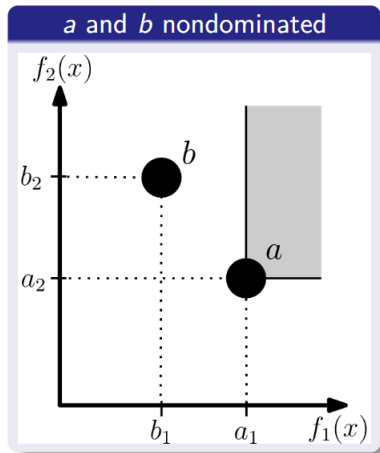
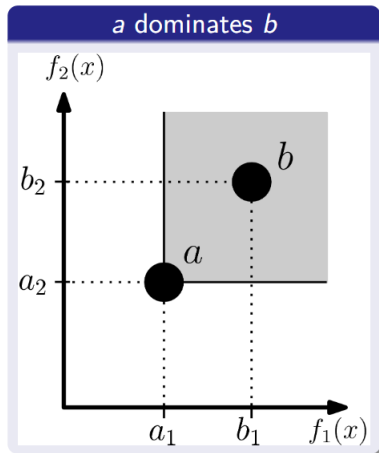
Objectives: Maximising efficiencies at different flow rates

Black Box Optimisation Problem



No analytical expression for objective functions are available

Comparing Solutions



Comparing Solutions

Which solutions are non dominated - both objectives are to be minimised

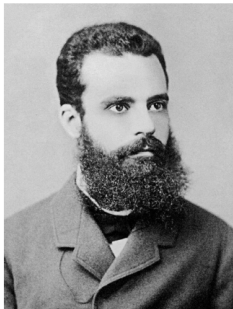
Solution	$f_1(x)$	$f_2(x)$
a	2	3
b	5	5
c	4	1
d	4	3

Options:

- ① a and b
- ② a and c
- ③ a, c and d
- ④ All of them

Pareto Optimality

Vilfredo Pareto



Unknown photo authorship (pre 1926)

Italian Polymath, Engineer and
Academic

A decision vector $\mathbf{x}$ is Pareto optimal iff $\nexists \mathbf{x}' \in \mathcal{X}$ where $\mathbf{x}' \prec \mathbf{x}$

$\mathbf{x}$ is therefore optimal in the sense that it is impossible to find an alternative solution that improves performance in any objective, without having to also degrade performance in at least one other objective

Pareto Set and Pareto Front

From Pareto optimality we can define the optimal set of solutions to the multi-objective optimisation task as:

$$\mathcal{P} = \{\mathbf{x} \in \mathcal{X} \mid \nexists \mathbf{x}' \in \mathcal{X}, \mathbf{x}' \prec \mathbf{x}\}$$

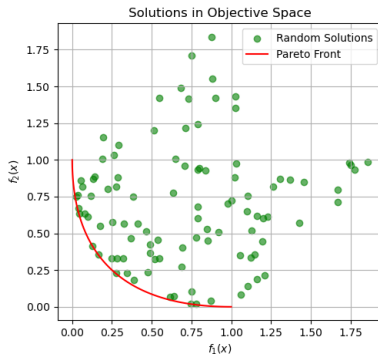
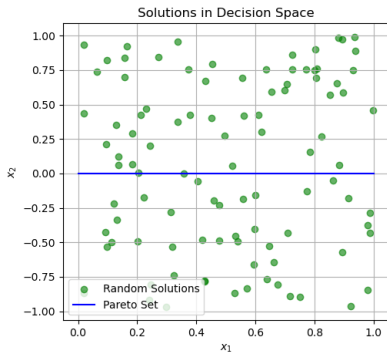
Note it is possible that the Pareto Set $\mathcal{P}$ for a problem to have infinite membership

The image of $\mathcal{P}$ under $\mathbf{f}$ is known as the Pareto front, $\mathcal{F}$

Pareto Set and Pareto Front

$$\begin{array}{ll}\text{Minimize} & f_1(x_1, x_2) = x_1^2 + x_2^2 \\ \text{Minimize} & f_2(x_1, x_2) = (x_1 - 1)^2 + x_2^2 \\ \text{Subject to} & 0 \leq x_1 \leq 1 \\ & x_2 = 0\end{array}$$

Pareto Set and Pareto Front



Summary So Far

- In multi-objective optimisation, we seek to optimise more than one objective simultaneously
- We are concerned with an approximation to the Pareto front and Pareto set - to define our trade-off solutions